

Data Dictionary: food_health.csv

Overview

This document provides a data dictionary for the food_health.csv. The dataset contains 656 variables derived from NHANES dietary recall, laboratory, examination, and questionnaire components, along with constructed variables.

Dietary intake (grams)

Cols 1–288: `gr_X` For each food item/group name `X` appearing in the column header, `gr_X` equals the grams of `X` consumed in the 24-hour recall.

Examples (not exhaustive): `gr_skim_milk`, `gr_whole_milk`, `gr_yogurt`, `gr_coffee`, `gr_soda`, `gr_broccoli`, `gr_white_bread`, etc.

Recall-level nutrient totals / summary fields

Col 293: `vitaminc` Total vitamin C intake from the 24-hour recall (units as in NHANES dietary totals; commonly mg).

Col 294: `alcohol` Total alcohol intake from the 24-hour recall (units as in NHANES dietary totals).

Col 295: `caffeine` Total caffeine intake from the 24-hour recall (units as in NHANES dietary totals; commonly mg).

Col 296: `cholesterol` Total dietary cholesterol intake from the 24-hour recall (units as in NHANES dietary totals; commonly mg).

Dietary indicators (consumed or not)

Cols 299–586: `X` For each food item/group name `X` appearing in the column header (same naming as the `gr_X` variables), `X` is a binary indicator equal to 1 if the respondent consumed `X` in the 24-hour recall and 0 otherwise.

Examples (not exhaustive): `skim_milk`, `whole_milk`, `yogurt`, `coffee`, `soda`, `broccoli`, `white_bread`, `energy_drink`, `sports_drink`, etc.

NHANES exam/questionnaire/lab variables

- Col 595:** `work_type` Employment type classification (constructed from NHANES occupation/employment questionnaire fields).
- Col 596:** `work_hs` Indicator for high-stress or physically demanding job (constructed).
- Col 597:** `work_occupation` Occupation category (derived from NHANES occupation codes).
- Col 601:** `gender` Biological sex indicator (male/female).
- Col 602:** `age` Age in years at time of examination.
- Col 603:** `race` Race/ethnicity category (NHANES standard categories).
- Col 604:** `edu` Education level category.
- Col 605:** `hh_size` Household size (number of persons in household).
- Col 606:** `hh_income` Household income category (survey-reported bracket).
- Col 607:** `indfmin2` Family income-to-poverty ratio (NHANES socioeconomic measure).
- Col 608:** `smq020` Smoking status indicator (e.g., smoked at least 100 cigarettes in lifetime).
- Col 609:** `smq040` Current smoking frequency (every day / some days / not at all).
- Col 610:** `smq050q` Quantity of cigarettes smoked (self-reported number).
- Col 611:** `smq050u` Unit of cigarette frequency reporting (e.g., per day, week, month).
- Col 617:** `paq635` Days per week of moderate recreational activity.
- Col 618:** `paq640` Minutes per day of moderate recreational activity.
- Col 619:** `paq650` Days per week of vigorous recreational activity.
- Col 620:** `paq655` Minutes per day of vigorous recreational activity.
- Col 621:** `pad660` Minutes per day spent sedentary (e.g., sitting).
- Col 622:** `dr1tcaff` NHANES Day 1 total caffeine (dietary totals variable; units per NHANES documentation).
- Col 623:** `dr2tcaff` NHANES Day 2 total caffeine (dietary totals variable; units per NHANES documentation).

Constructed variables

- Col 625: veggies** Indicator/aggregate for vegetable consumption (constructed from dietary variables).
- Col 626: fruits** Indicator/aggregate for fruit consumption (constructed from dietary variables).
- Col 627: veggies_n_fruits** Combined indicator/aggregate for fruit and vegetable consumption.
- Col 628: veggies_gr** Total grams of vegetables (constructed aggregate).
- Col 629: fruits_gr** Total grams of fruits (constructed aggregate).
- Col 630: veggies_n_fruits_gr** Total grams of fruits + vegetables (constructed aggregate).
- Col 631: coffee_espresso_unit** Processed coffee measure (espresso-unit scaling; constructed for teaching use).
- Col 632: beef_all** Total beef consumption (constructed aggregate across beef-related food groups).
- Col 633: red_meat_all** Total red meat consumption (constructed aggregate across beef/pork/lamb/etc.).
- Col 634: nuts** Total nuts consumption (constructed aggregate across nut-related groups).
- Col 635: age2** Age-squared (constructed).
- Col 636: age_cat** Age category (constructed).
- Col 637: married** Marital status indicator (constructed from NHANES marital status fields).
- Col 638: hh_income_usd** Household income in USD (constructed/cleaned).
- Col 639: hh_income_percap** Household income per capita (constructed).
- Col 640: ln_hh_income_percap** Log household income per capita (constructed).
- Col 641: income_cat** Income category (constructed).
- Col 642: sport_days** Days of sports/exercise (constructed from PAQ/PAD fields).
- Col 643: walk_cycle_days** Days walking/cycling (constructed from PAQ/PAD fields).
- Col 644: smoker** Smoking indicator (constructed from SMQ fields).
- Col 645: sleep_hs** Hours of sleep (constructed/cleaned).

Potential Outcomes (cleaned/constructed)

- Col 646:** `bp_systolic` Systolic blood pressure (mmHg; cleaned/selected reading).
- Col 647:** `bp_diastolic` Diastolic blood pressure (mmHg; cleaned/selected reading).
- Col 648:** `total_cholesterol` Total serum cholesterol (mg/dL; lab).
- Col 649:** `hdl` HDL cholesterol (mg/dL; lab).
- Col 650:** `weight` Body weight (kg; exam).
- Col 651:** `height` Height (cm; exam).
- Col 652:** `ldl` LDL cholesterol (mg/dL; lab).
- Col 653:** `bmi` BMI (kg/m^2 ; cleaned/duplicated from `bmxbmi`).
- Col 654:** `normal_weight` Indicator for normal BMI range (constructed).
- Col 655:** `blood_pressure` Constructed blood pressure index (in this file: `bp_systolic` + `bp_diastolic`).
- Col 656:** `heart_risk` Constructed cardiovascular risk index (composite of BP and lipid measures; defined in the class materials / data-prep step).

Exam and Lab variables (PROBABLY IGNORE)

- Col 587:** `bmxbmi` Body mass index from NHANES exam file (BMI; kg/m^2).
- Col 588:** `peascst1` Physical activity status indicator (constructed from NHANES physical activity questionnaire items; reflects engagement in moderate/vigorous activity).
- Col 589:** `bpq150a` Self-reported hypertension diagnosis indicator (e.g., ever told by a health professional of high blood pressure).
- Col 590:** `bpq150b` Follow-up hypertension questionnaire item (e.g., current treatment or monitoring).
- Col 591:** `bpq150c` Additional hypertension-related questionnaire item.
- Col 592:** `bpq150d` Additional hypertension-related questionnaire item.
- Col 593:** `bpxsy2` Second systolic blood pressure reading (mmHg) from NHANES examination component.
- Col 594:** `bpxdi2` Second diastolic blood pressure reading (mmHg) from NHANES examination component.
- Col 598:** `cdq001` Cardiovascular disease questionnaire item (e.g., coronary heart disease diagnosis).
- Col 599:** `cdq002` Cardiovascular disease questionnaire item (e.g., angina or heart attack diagnosis).

Col 600: `cdq010` Cardiovascular disease questionnaire item (e.g., stroke diagnosis).

Col 612: `lbdtcsl` Total serum cholesterol (SI units).

Col 613: `lbttr` Triglycerides (conventional units).

Col 614: `lbdtrsl` Triglycerides (SI units).

Col 615: `lbdldl` LDL cholesterol (conventional units).

Col 616: `lbdldlsi` LDL cholesterol (SI units).

Identifiers, recall-item fields (IGNORE)

Col 289: `seqn` NHANES respondent identifier (SEQN).

Col 290: `food_code` NHANES food code (8-digit code) associated with the item referenced in `grams`.

Col 291: `days` Recall day indicator (typically 1 = Day 1 dietary recall, 2 = Day 2 dietary recall).

Col 292: `grams` Grams consumed of the food item indicated by `food_code` on the recall day.

Col 297: `year` Survey cycle/year label used in the processed dataset (e.g., NHANES cycle indicator).

Col 298: `gramsm` Processed summary measure from the original data-prep pipeline (interpret as an alternative scaling/summary of recall intake).

Col 624: `_merge` Merge indicator from the data construction step (e.g., matched across NHANES components).